

Maubach Subdivision on $\mathbb{CP}^2$

Related Work

Vinicius Mello

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Abstract

This document surveys the literature related to a planned article built on the technical report *Maubach Subdivision on $\mathbb{CP}^2$* . The article has three strands: (i) the combinatorial description of Maubach’s subdivision as a *stellar subdivision scheme* [42] and its adaptation to arbitrary compatible initial meshes; (ii) pointerless LPT codes [5] and their extension to non-cubical seeds; and (iii) the triangulation of Riemann surfaces as codimension-2 loci of $\mathbb{CP}^2$ as an application. For each strand we list the closest prior work and state precisely how our approach compares. Along the way we confirm computationally that Gaifullin’s 15-vertex triangulation of $\mathbb{CP}^2$ [25] is *balanced*, i.e. properly 5-colorable, as Gaifullin himself states (with minimality among balanced triangulations of $\mathbb{CP}^2$, [25, §2]). Its Maubach compatibility is therefore an instance of the classical “coloring $\Rightarrow$ reflected” mechanism. Items marked [verify] were not checked against the original source.

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1 Summary and positioning

1. **The direct competitor is Dienen, Gehring and Storn (DGS) [20].** They initialize Maubach’s routine on *any* conforming triangulation in any dimension by a proper $(N+1)$ -coloring of the vertices, $N \geq n$. The coloring is computed greedily in linear time, with N bounded by the maximal vertex degree. For $N = n$ the coloring is exactly the “global flag/rank” mechanism behind Theorems 9 and 10 of the thesis. For $N > n$ each n -simplex

is treated as a face of a virtual N -simplex. The article must position itself explicitly with respect to DGS.

2. **Gaifullin’s $\mathbb{CP}^2$ is balanced** (Proposition 1). The Maubach ordering found by the search of the report is precisely a proper 5-coloring of the vertices. This contradicts the report’s statement that the mesh has no global rank. What is true is that the rank is not given *by construction*. This yields a clean statement for the article: balanced $\Rightarrow$ Maubach-compatible with $i = j$ on every shared facet [54, 20].
3. **Kühnel’s $\mathbb{CP}_9^2$ cannot be balanced**. It is 3-neighborly, so its 1-skeleton is K_9 and needs 9 colors. In DGS terms it is the case $N = 8 > n = 4$.
4. **Theorems 9 and 10 of the thesis are not new as facts**. Products of simplices and barycentric subdivisions are the textbook balanced complexes, and “balanced $\Rightarrow$ reflected” is already in Traxler [54]. The pre-refinement of Kossaczky [34] and Stevenson [53, App. A] is essentially the barycentric subdivision without the edge-midpoint level, with $(n+1)!/2$ cells instead of $(n+1)!$. What belongs to the thesis is the *formalism*:
 - a scheme is an invariant plus an algorithm;
 - the invariant is proved inductively, level by level (Lemmas 5–7, Theorem 8);
 - Mitchell’s facet-based and Maubach’s edge-based schemes are unified in one stellar language.
5. **The LPT strand has strong analogues but no equivalent**. Existing work includes t8code/TM-index [16], concurrent binary trees [22], diamond hierarchies [59] and permutahedral encodings [13]. None of them combines conforming bisection, arbitrary dimension, a non-cubical seed and no pointers.
6. **The application occupies a real gap**. We triangulate the curve *intrinsically in $\mathbb{CP}^2$* : compact, with no charts, no branch cuts and no special treatment of points at infinity. We do this by combinatorial continuation over a Maubach-refined mesh of $\mathbb{CP}^2$ itself. The weakness to state openly is that the output is *not* topologically certified.

2 Bisection refinement: newest vertex and Maubach

2.1 The classical lineage

Work	Contribution	Requirement on the initial mesh
Rivara [50, 51]	Longest-edge bisection (2D/3D/ n D)	None combinatorial; the choice is geometric
Mitchell [43, 44]	Newest vertex bisection (NVB), 2D	Compatible refinement edges; always exist in 2D [11]
Bänsch [6], Kossaczky [34] Maubach [40] Traxler [54]	NVB in 3D n D NVB: bisect $[v_0, v_\gamma]$, cyclic tag Equivalent formulation	Kossaczky pre-refines the initial mesh Grids “generated by reflection” Reflected partition. For simply connected domains: exists iff every interior $(n-2)$ -face lies in an even number of simplices
Arnold–Mukherjee– Pouly [4]	Marked tetrahedral bisection (5 types)	Any 3D mesh, but only uniform refinements of level $\equiv 0 \pmod{n}$ are conforming
Stevenson [53]	Closure estimate in n D	Matching-neighbour condition (b), weaker than Traxler’s. Open whether every 3D mesh satisfies it. App. A: pre-refinement into $\frac{1}{2}(n+1)!$ simplices always does
Alkämper–Gaspoz– Klöforn [1]	Weak compatibility: $O(n)$ renumbering of any mesh. Termination, not optimal closure	Any
Belda-Ferrín et al. [7, 9, 8]	n marked bisection stages, then “cast” to reflected simplices and switch to Maubach	Any conforming mesh
Diening–Gehring– Storn [20]	Initialization by $(N+1)$ -coloring; greedy, $O(\#\text{vertices})$; shape regularity and BDV closure	Any. $N = n$ is Traxler’s condition
Diening–Storn– Tscherpel [21]	Optimal grading; vertex generation from colors	$(n+1)$ -colored
Gehring [26]	Constant in the BDV–Stevenson theorem	–
Li [38]	First unconditional closure estimate for AMP/Bänsch	Any (3D)

Belda-Ferrín et al. state that, for $n > 3$, “there is not any known method to extract a reflection structure from an arbitrary unstructured conformal mesh” [8]. DGS [20] closes this gap by coloring, without extracting a reflection structure. Their Remark 9 notes that $(n+1)$ -colorings do not always exist. Their Section 3.2 therefore allows $N > n$ colors: each simplex is embedded as a face of a virtual N -simplex, and the bisection vertex generations are propagated accordingly. Their Theorem 10 shows that the resulting routine coincides with Maubach’s original one under the ordering of their Definition 2.

2.2 Where the thesis fits

Definition 9 versus Stevenson’s condition (b). (MaubachA) allows two same-level neighbours to have either $i = j$ or $\{i, j\} = \{0, \text{type}\}$. In Maubach’s convention (bisection edge $[0, \text{type}]$) this is the analogue of Stevenson’s use of

$$T_R = (x_n, x_1, \dots, x_\gamma, x_{n-1}, \dots, x_{\gamma+1}, x_0)_\gamma,$$

the tagged simplex with the same children as T , in Traxler’s convention (bisection edge x_0x_n). There, T and T' are *reflected neighbours* if the vertex sequence of T or of T_R agrees with that of T' in all but one position [53, §4]. [verify] the exact equivalence after translating conventions.

The structural difference is this. The thesis states an invariant that holds at *every* level and proves that the algorithm preserves it (Lemma 7, Theorem 8). Stevenson and Traxler impose their condition on the initial mesh only, and then prove that all uniform refinements are conforming (Stevenson’s Theorem 4.3). Both routes prove essentially the same theorem; the thesis’s proof is local, inductive and purely combinatorial. Stevenson’s Remark 4.4 shows that condition (b) is also *necessary*. Definition 9 can thus be presented as a hereditary form of the weakest possible level-0 condition.

Theorems 9 and 10. Both constructions are balanced complexes in Stanley’s sense. For a product of simplices, the lattice rank $|L|$ is an $(n+1)$ -coloring; for a barycentric subdivision, the dimension of the flag element is. “Balanced $\Rightarrow$ reflected with $i = j$ ” is Traxler’s condition, restated as a coloring in DGS §3.1. The thesis (2006) predates Stevenson (2008) but not Traxler (1997), so priority on the general fact cannot be claimed. Theorems 9 and 10 are best presented as instances with short proofs in the stellar formalism. Stevenson’s pre-refinement costs half as many cells as the barycentric one, and DGS show that neither is needed. For non-balanced meshes the fair comparison is therefore DGS with $N > n$.

Where there is room for a genuine contribution.

1. *Closed manifolds as seeds.* All work cited above concerns domains in $\mathbb{R}^n$, for finite elements. Triangulations of closed manifolds ($\mathbb{CP}^2$, $S^2 \times S^2$) and the question of which minimal triangulations are balanced belong to combinatorial topology [32, 31, 57]. No bridge to NVB seems to exist.
2. *Gaifullin’s triangulation is balanced* (Proposition 1). *Correction:* this is stated by Gaifullin himself, who also proves that his triangulation is vertex-minimal among balanced (“regularly vertex-coloured”) triangulations of $\mathbb{CP}^2$ [25, §2 and Prop. 2.3]. The novelty is only the connection with Maubach bisection.
3. *Stellar subdivision schemes as a unifying language.* Mitchell (facet) and Maubach (edge) fit the same form. This connects to Alexander’s and Newman’s stellar moves [39], to Velho’s stellar grammars [55] and 4–8 meshes [56] (which are NVB in 2D), and to balanced cross-flips [31], the balanced analogue of Pachner moves.

2.3 Colorability criteria

Joswig [32] shows that a locally strongly connected simplicial complex is balanced if and only if its group of projectivities is trivial. For simply connected manifolds this reduces to Traxler’s criterion: every $(n-2)$ -face has even degree. Two elementary facts suffice for the $\mathbb{CP}^2$ seeds:

- a k -neighborly complex with $k \geq 2$ and more than $n + 1$ vertices is never balanced, since adjacent vertices need distinct colors;
- $\mathbb{CP}^2$ is simply connected, so it suffices to check that every triangle lies in an even number of 4-simplices.

Proposition 1. *Gaifullin’s 15-vertex, 108-cell triangulation of $\mathbb{CP}^2$ is balanced. Explicitly, its vertices split into five classes of three mutually non-adjacent vertices. The complement of the 1-skeleton is a disjoint union of five triangles, and every vertex has degree 12. In the ordering used by `riemann_cp2`, the position of each vertex is the same in every cell containing it. Hence all 270 interior facets satisfy $\partial_i\sigma = \partial_j\omega \Rightarrow i = j$. Moreover, each of the 240 triangles lies in*

4 (180 triangles) or 6 (60 triangles) cells, and the dual graph is bipartite. Kühnel’s $\mathbb{CP}_9^2$ is not balanced.

Verification. Direct computation on the facet list `gaifullin_cells_raw` in `examples/top/riemann_cp2/riemann_cp2.cpp`:

- exhaustive backtracking finds a proper 5-coloring;
- the per-vertex set of positions is a singleton for all 15 vertices;
- the facet-index pairs are (k, k) , 54 times for each $k = 0, \dots, 4$;
- triangle degrees were counted directly.

For Kühnel’s $\mathbb{CP}_9^2$, 3-neighborliness gives a K_9 skeleton. □

Remark. *Gaifullin also shows that the automorphism group $S_4 \times S_3$ acts by Fubini–Study isometries [25]. This explains the measured max/min Fubini–Study edge ratio of exactly 1.4636 with zero variance across all 108 cells. The same paper gives a 33-vertex subdivision on which the moment map is simplicial, a candidate seed worth testing.*

3 Pointerless representations and neighbour finding

Work	Hierarchy / idea	Seed	Conf.	Dim.
Hebert [30]	Symbolic tetrahedral bisection	Cube	yes	3
Maubach [41]	Neighbour location in n D Maubach grids	Kuhn	yes	n
Evans–Kirkpatrick–Townsend [24]	Right-triangulated irregular networks: binary label $\Rightarrow$ geometry	Square	yes	2
Lee–De	Constant-time neighbours for bisection	Cube (6 tets)	yes	3
Floriani–Samet [37]	tetrahedra			
Atalay–Mount [5]	LPT codes: geometry and neighbours by table lookup	Kuhn cube (Sym(n) orbit)	yes	n
Weiss–De	Diamond hierarchies (diamond = star of the bisection edge)	Kuhn	yes	n
Floriani [59]				
Burstedde–Holke [16], t8code	TM-index (Morton SFC for Bey red refinement); <i>forest</i> over an arbitrary coarse mesh	Any	no	2, 3
Burstedde–Wilcox–Ghattas [17]	Forest of octrees	Any (hex)	no	2, 3
Dupuy [22]	Concurrent binary trees: implicit heap of bits; longest-edge bisection on GPU	2D	yes	2
Boissonnat–Kachanovich–Wintraecken [13]	Permutahedral encoding of Freudenthal–Kuhn/Coxeter simplices	$\mathbb{R}^d$ (uniform)	yes	d

Positioning of glpt.

- *Against Atalay–Mount.* Their same-orthant neighbour formula relies on the seed being a single Sym(n) orbit. A *balanced* seed suffices. Ordering the vertices of each seed cell by color identifies it affinely with the Kuhn reference simplex. The LPT code inside a seed is then unchanged, and crossing a seed boundary preserves the local facet index ($i = j$). Such crossings are resolved by a seed adjacency table, 108×5 for Gaifullin. This explains *why* the adaptation is easy. It is less easy in two cases:

- seeds compatible only through the $\{0, \text{type}\}$ exception would need a permutation correction at crossings;

- DGS seeds with $N > n$ would need codes of a virtual N -simplex, and DGS’s Remark 12 warns that restricting refinements of the virtual N -simplex does *not* reproduce their refinement.

Both are future work.

- *Against t8code.* The architecture is the same (tree id $\approx$ `seed_index`, per-tree linear code, inter-tree connectivity table). Ours uses *conforming*, graded bisection without hanging nodes, in arbitrary dimension (4 here).
- *Against BKW.* They encode a uniform triangulation; `glpt` encodes an adaptive hierarchy.
- *Limitations to state.* The `DEEP_CROSSING` case (neighbour refined deeper than the query) is not implemented. Storage is $O(\#\text{leaves})$ in a hash table.

4 Application: triangulating Riemann surfaces

4.1 Numerical computation of Riemann surfaces

Deconinck and van Hoeij’s Maple package `algcures` [19] computes monodromy and period matrices of plane algebraic curves. Other tools in the same vein include Frauendiener–Klein’s Matlab codes, Sage’s `RiemannSurface` (Bruin–Sijlsing–Zotine), and Kranich’s certified homotopy continuation of plane curves [35]. See Bobenko–Klein [12] for an overview. These methods treat the surface as a branched cover of $\mathbb{CP}^1$ and compute analytic invariants. We construct a *simplicial complex* of the curve in $\mathbb{CP}^2$, which is complementary: our mesh can, for instance, confirm the genus through $\chi = 2 - 2g$.

4.2 Visualization

Hanson [29] renders Fermat curves in $\mathbb{CP}^2$ projected to $\mathbb{R}^3$, using parametrizations and symmetry. Nieser–Poelke–Polthier [46] build meshes as multivalued graphs over $\mathbb{C}$ from branch points, with cuts. Kranich renders domain-coloured Riemann surfaces on the GPU [35]. All of these work through a projection to $\mathbb{C}$, with charts or cuts. Our construction is intrinsic in $\mathbb{CP}^2$; the price is the chart-consistency problem at visualization time.

4.3 Implicit manifolds of codimension greater than one

- **PL continuation.** Allgower–Schmidt [3] approximate implicit manifolds of arbitrary codimension over Freudenthal–Kuhn triangulations. The textbook treatment of simplicial continuation is Allgower–Georg [2]. This is the historical lineage of our “continuation + Kuhn”.
- **Weigle–Banks** [58], complex-valued contour meshing, treat the zero set of $f : \mathbb{C}^2 \rightarrow \mathbb{C}$ as a surface in $\mathbb{R}^4$, extracted simplicially on a grid. This is the most direct predecessor of our codimension-2 extraction, but it works on a uniform affine grid of $\mathbb{R}^4$, not on $\mathbb{CP}^2$.
- Bhaniramka–Wenger–Crawfis [10] construct isosurfaces in any dimension, in codimension 1.
- Castelo et al. [18], the J1a triangulation: adaptive, any dimension, for implicit manifolds. This is the Brazilian lineage of [28] and a natural competitor on nD adaptivity.
- **Boissonnat–Kachanovich–Wintraecken** [13] trace isomanifolds from a seed point by cell-to-cell continuation over Freudenthal–Kuhn/Coxeter triangulations, polynomially in d and with guarantees; see also [14] for topological correctness. This is a mandatory reference: same strategy, uniform triangulation, certified. Ours is adaptive and lives in $\mathbb{CP}^2$, but is not certified.

- Ju et al. [33] build adaptive simplicial grids for implicit complexes, including curve networks (codimension 2 in $\mathbb{R}^3$), starting from a 6-tetrahedron cube. [verify] their exact bisection rule and refinement criterion.
- Gerstner–Rumpf [27] and Pascucci [47] extract isosurfaces on bisection hierarchies. The combination “Maubach hierarchy + extraction” is thus established in visualization, but only in codimension 1 and in $\mathbb{R}^3$.

4.4 Certification

Plantinga–Vegter [48] guarantee isotopy via interval arithmetic on quadrees/octrees. Mourrain–Pavone [45] subdivide in the Bernstein basis. Reuter et al. [49] use the *barycentric* Bernstein basis on simplices, the exact counterpart of our `-bernstein` enclosure on 2-faces. Bertini_real [15] computes numerical cell decompositions of real curves and surfaces; a complex curve is a real surface in $\mathbb{R}^4$ ([verify] whether it handles this case). Our method is heuristic and adaptive. `-tangency-threshold` and `-repair-rounds` reduce the bad-cell rate (best: 0.180%) but do not eliminate it. It should be presented as a practical triangulation, checkable a posteriori via $\chi = 2 - 2g$, not as a certified one.

5 Triangulations of $\mathbb{CP}^2$

- Kühnel–Bachhoff [36]: $\mathbb{CP}_9^2$, the unique 9-vertex triangulation. It is 3-neighborly, hence not balanced.
- Gaifullin [25]: 15 vertices, automorphism group $S_4 \times S_3$ acting by Fubini–Study isometries; minimal among triangulations admitting a chess colouring. Balanced by Proposition 1. Note: the technical report cites arXiv:0904.4222 with an incorrect title; the correct title is the one given in the bibliography.
- Schwartz [52]: trisection of $\mathbb{CP}_9^2$.
- Tools used for verification: GAP with simpcomp [23].

6 Suggestions for the article

1. **Central statement.** “Maubach bisection works verbatim on any balanced initial triangulation, including triangulations of closed manifolds; pointerless LPT codes extend to such seeds by adding a seed index and a seed adjacency table.” The first half is known [54, 20]. The contributions are:

- the star-scheme formalism with a hereditary invariant;
- closed-manifold seeds (Gaifullin’s balanced $\mathbb{CP}^2$);
- the LPT extension;
- the application.

2. **Comparison experiments.**

- Kühnel with DGS initialization ($N = 8$) against Gaifullin ($N = 4$): cells, Fubini–Study quality, bad-cell rate. This would be the first use of DGS outside $\mathbb{R}^n$.
- Barycentric subdivision of Kühnel against Stevenson’s pre-refinement of Kühnel against Gaifullin: cell cost and quality.
- A Weigle–Banks style extraction on an affine chart grid against intrinsic $\mathbb{CP}^2$ on the same curve: cells visited and artifacts near infinity.

3. **Corrections to the report** before reusing text:

- Gaifullin *has* a global rank structure (a coloring);
- fix the title of the Gaifullin reference;
- reinterpret the report’s “coloring problem on the dual graph” as a search for a proper $(n+1)$ -coloring of the vertices.

4. **Open questions.** What is the minimal number of vertices of a balanced triangulation of $\mathbb{CP}^2$? Does Gaifullin’s 33-vertex subdivision (simplicial moment map) give a better seed?

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